

Zephyr OS

An open-source RTOS ecosystem for secure, connected IoT solutions

Clark Jarvis (MCU Software and Tools Product Marketer)

October 2026

restricted

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3 Getting started with Zephyr (Development Board Support)

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Confirmation of prework

- Zephyr needs eight command-line tools on Windows - a prework script installs all of them
  - **cmake, ninja, python, git, wget, 7z, gperf, dtc**

- Run from a regular, non-admin PowerShell window:
  - **powershell -ExecutionPolicy Bypass -File .\setup-zephyr-windows-deps.ps1**

- You're done when verification shows eight **[OK] lines**

- Debug probe software - already installed in this room for ModusToolbox™ classes:
  - KitProg3 boards (CY8CPROTO-041TP, KIT_PSE84_EVAL) → **Infineon's OpenOCD**
  - KIT_PSC3M6_EVAL → onboard **SEGGER J-Link (needs V9.78 or later)**

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Setup Steps - What You'll Need Before You Build

- Every attendee leaves this session with a working Zephyr workspace, a toolchain, and your own board blinking

- One scheduling fact shapes everything below:
  - Setting up a Zephyr workspace pulls about 425 MB over the network and lands as roughly 2.5 GB on disk
  - Start the download in the first few minutes of this session

- You leave with:
  - A workspace holding the training fork of Zephyr and every lab application you need this week
  - A toolchain your builds can find, and a programmer that can reach your board
  - One blinky, built by you, flashed by you, running on your own board

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First Thing in the Room - Start the Download

- Install west, Zephyr's command-line tool:
  - **pip install west**
- Create the workspace and start the long pole downloading:
  - **west init -m https://github.com/ClarkJ-Infineon/devcon-training-2026 devcon-ws**
  - **cd devcon-ws**
  - **west update --narrow**
- --narrow is not optional - without it west fetches every branch/tag instead of the one pinned commit
- ~280 MB over the network, landing as ~1.25 GB on disk - leave it running and go listen to the presentation
- Full copy-paste commands: the manifest repo's README.md -> Quick start (same URL, viewable on GitHub)

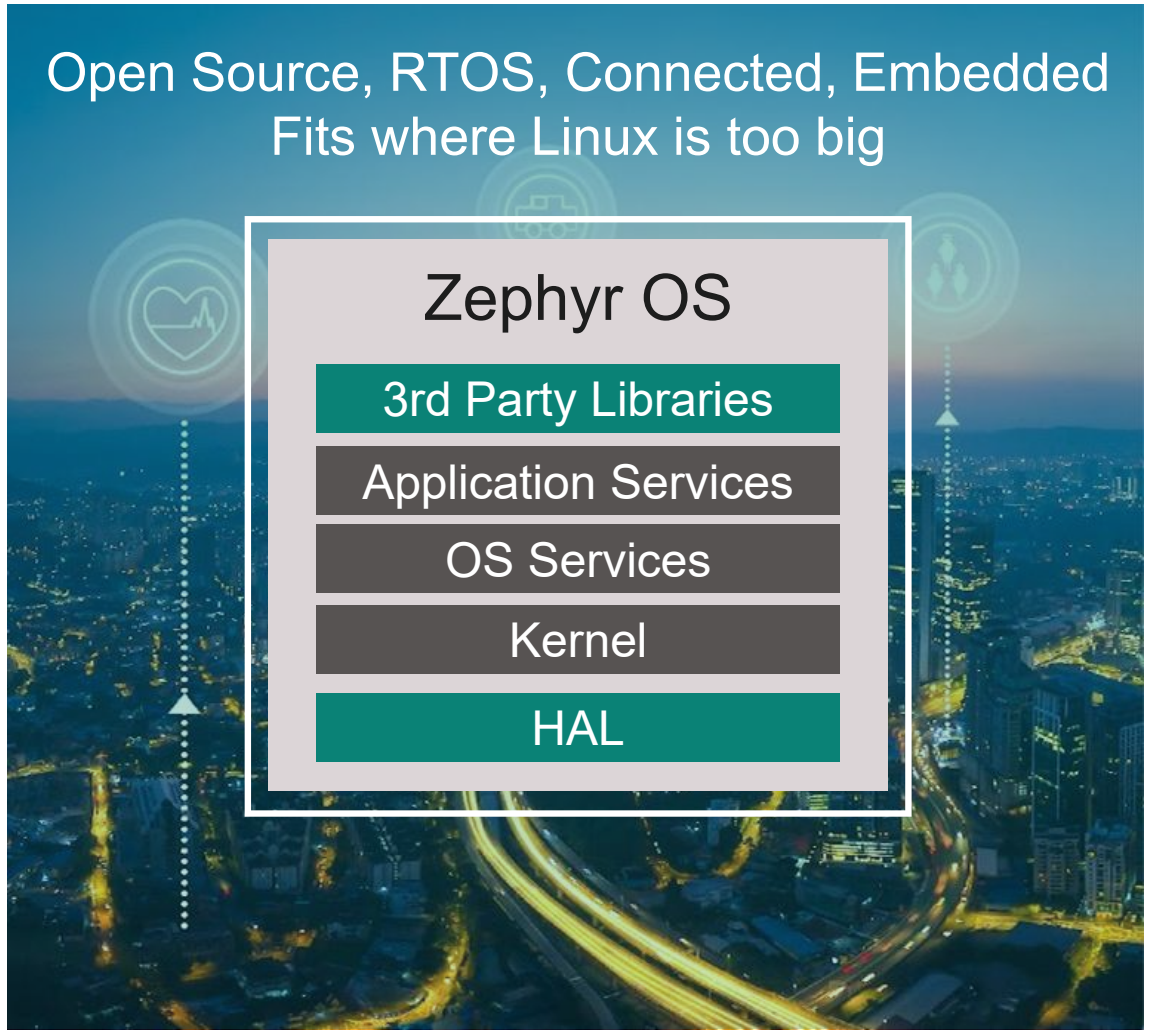
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Zephyr Project Overview

Content in this section is taken from Zephyr marketing collateral

Zephyr Project

- Open-source real time operating system
- Developer friendly with vibrant community participation
- Built with safety and security in mind
- Broad SoC, board and sensor support.
- Vendor Neutral governance
- Permissively licensed - Apache 2.0
- Complete, fully integrated, highly configurable, modular for flexibility
- Product development ready using LTS includes security updates
- Certification ready with Zephyr Auditable



THE LINUX FOUNDATION PROJECTS

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Use cases for a real-time OS



Industrial IoT



Asset Tracking



Wearables



Automotive



Healthcare



Worker Safety

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Products Running Zephyr Today



Oticon More Hearing Aid



Lildog & Lilcat
Pet Tracker



Livestock Tracker



Moto Watch 100



Samsung Galaxy Ring



Proglove



Adhoc Smart Waste



Google Chromebook



Framework laptop



Keeb.io BDN9



Hati-ACE



Blackhole™
PCIe AI Accelerator



BLiXT solid state circuit
breaker



Aethero Deimos Satellite



PHYTEC Distancer



Laird Connectivity sensors
& gateways



BeST pump monitoring



Vestas Wind Turbines

zephyrproject.org/products-running-zephyr

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1000+ supported boards… and growing

Infineon Boards



PSOC™ Control C3M5 Evaluation Kit



PSOC™ Edge E84 Evaluation Kit



PSOC™ 6 AI Kit



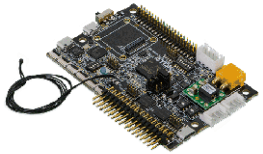
nRF9160 DK



STM32F746G Disco



M5StickC PLUS



TDK RoboKit 1



Arduino Portenta H7



ESP32



Arduino Nano 33 BLE



Intel UP Squared



Dragino LSN50 LoRA Sensor Node



Microchip SAM E54 Xplained Pro Evaluation Kit



Raspberry Pi Pico



Altera MAX10



NXP i.MX8MP EVK



Adafruit Feather M0 LoRa



u-blox EVK-NINA-B3

docs.zephyrproject.org/latest/boards

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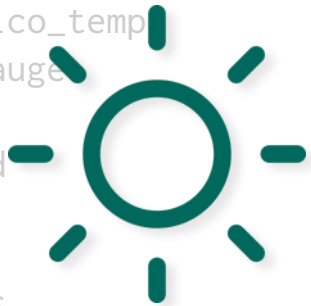
270+ Sensors Already Integrated

adt7420
adxl345
adxl362
adxl372
ak8975
amg88xx
ams_as5600
ams_iAQcore
apds9960
bma280
bmc150_magn
bme280
bme680
bmg160
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ccs811
dht
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qdec_sam
qdec_stm32
rpi_pico_temp
sbs_gauge
sgp40
sht3xd
sht4x
shtcx
si7006
si7055
si7060
si7210
sm351lt
stm32_temp
stm32_vbat
stmemsc
stts751
sx9500
th02
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ti_hdc20xx
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tmp108
tmp112
tmp116
vcnl4040
vl53l0x
wsen_hids
wsen_itds





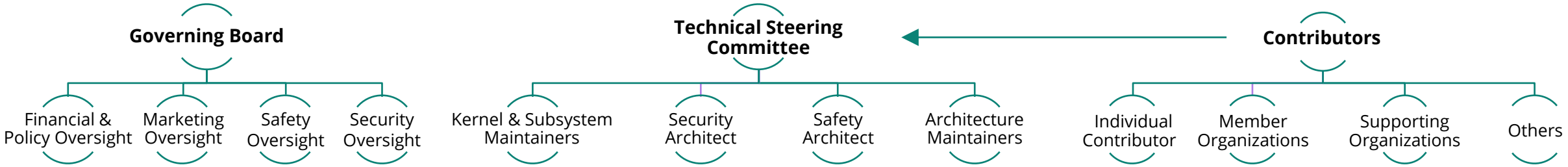


github.com/zephyrproject-rtos/zephyr/tree/main/drivers/sensor

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Zephyr Project Governance



Goal: Separate business decisions from meritocracy-based, technical decisions

Governing Board

- Decides project goals and strategic objectives
- Makes business , marketing and legal decisions
- Prioritizes investments and oversees budget
- Oversees marketing such as PR/AR, branding, others
- Identifies member requirements

Technical Steering Committee

- Highest technical decision body consisting of project maintainers and voting members
- Sets technical direction for the project
- Coordinates X-community collaboration
  - Sets up new projects
  - Coordinates releases
  - Enforces development processes
  - Moderates working groups
- Oversees relationships with other relevant projects

Community

- Code base is open to all contributors, need not be a member to contribute.
- Path to committer and maintainer status through peer assessed merit of contributions and code reviews
- Ecosystem enablement

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Zephyr Ecosystem

Content in this section is taken from Zephyr marketing collateral

Zephyr Ecosystem and Architecture

Governing Board, Technical Steering Committee, Contributors

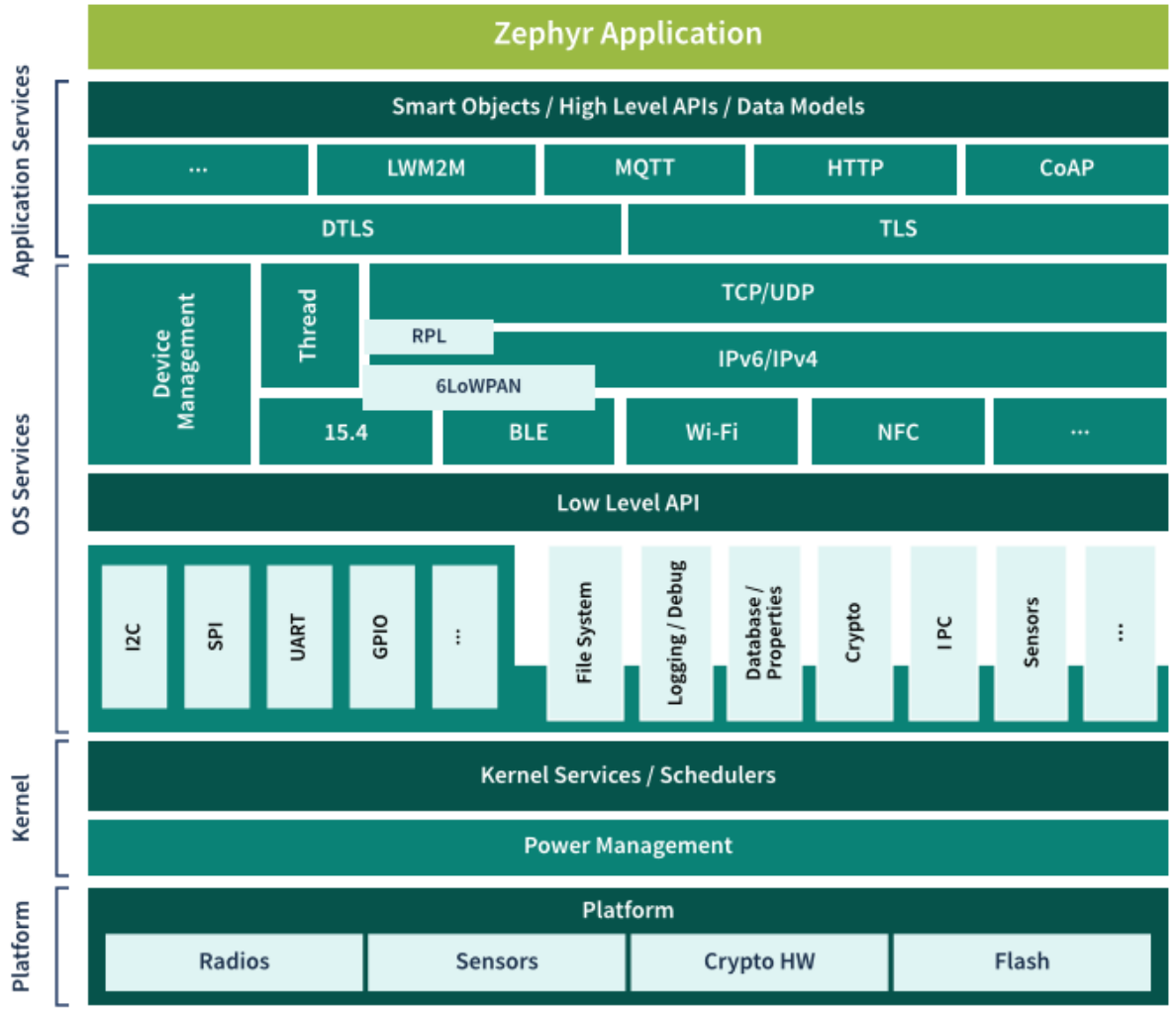
Applications & Middleware

Development Tools

Firmware & Libraries

Training & Consulting

Zephyr®



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Devicetree

- **Describe** & **configure** the available hardware on the target system
- **Decouple** the application from the hardware
- **Hardware config lives in Devicetree; software config lives in Kconfig**

```
zephyr,user {
    io-channels = <&adc0 12>;
};

aliases {
    cmd-pwm = &pwm_cmd;
};

pwmleds {
    compatible = "pwm-leds";

    pwm_cmd: pwm_cmd_led {
        label = "Command PWM (P5.0)";
        pwms = <&pwm0_6 0 PWM_USEC(1000)>;
    };
};
```

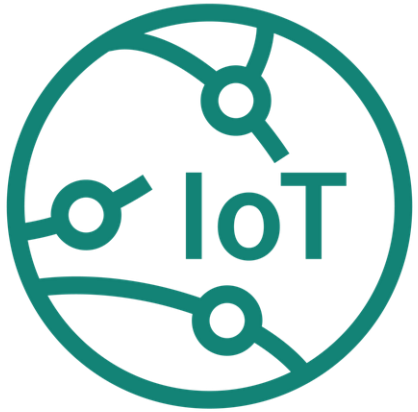


.overlay file example
(KIT_PSC3M6_EVAL)

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IoT Connectivity Options

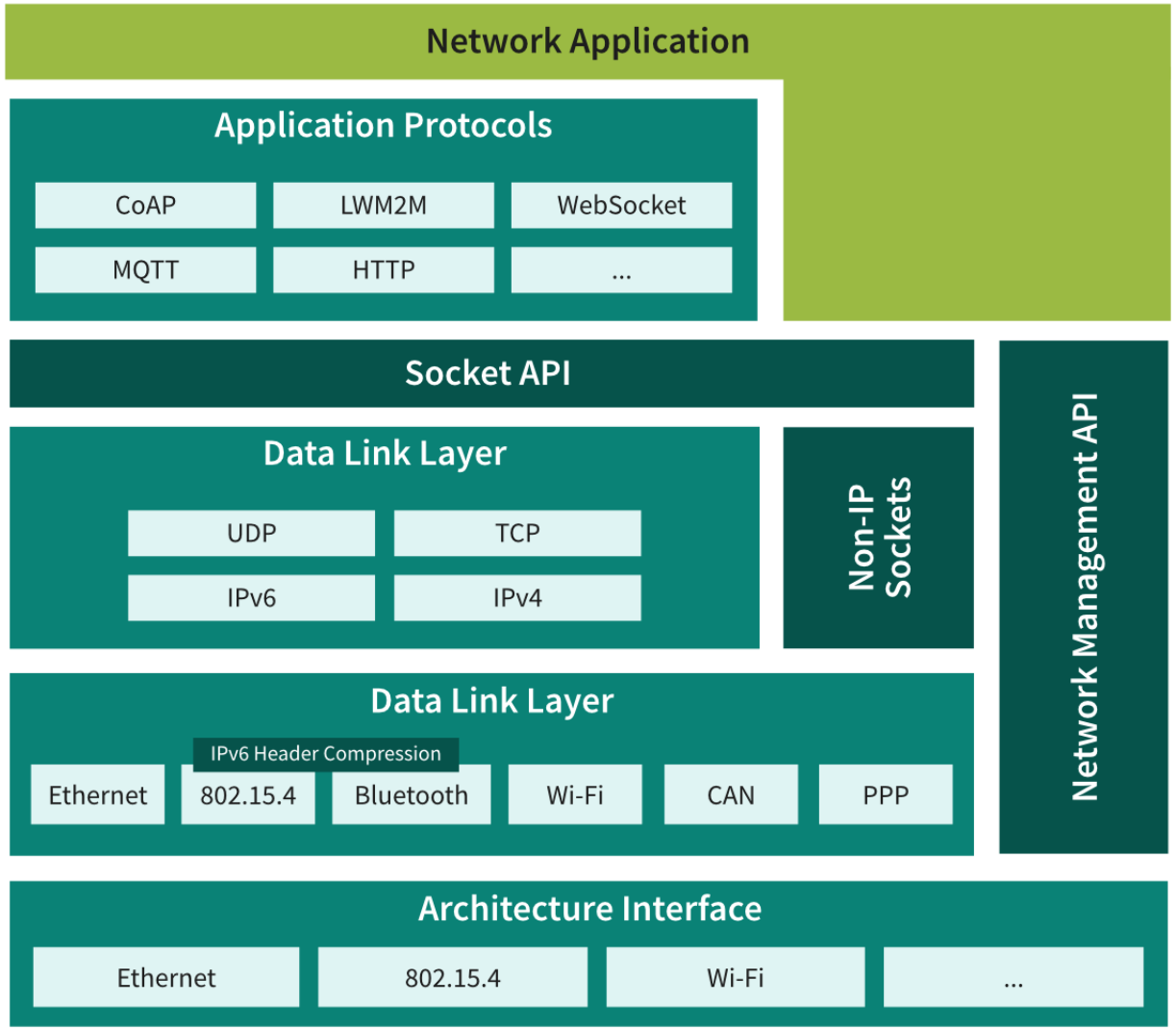
- Wide variety of **communication protocols**
  - Ethernet, 802.15.4, Thread, LoRa, Bluetooth, CAN bus, …
- **Core network protocols** like IPv6, IPv4, UDP, TCP, ICMPv4, and ICMPv6.
- **Security** (ex. TLS, DTLS, …)
- **Cloud integration** using MQTT, CoAP and HTTP protocols
- **Over-the-air updates**
- **Device management** using OMA LwM2M 1.1 protocol



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Native IP Stack

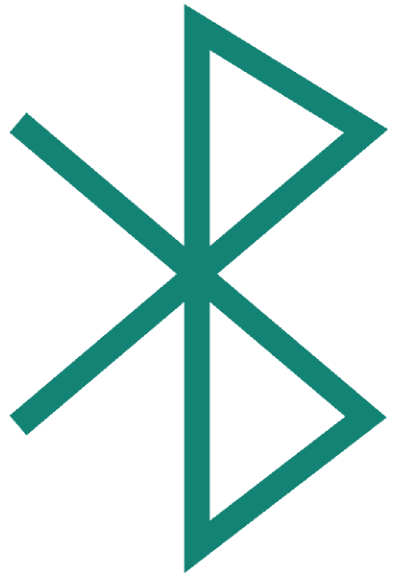
- Built from scratch, on top of Zephyr native kernel concepts
- Dual mode **IPv4/IPv6 stack**
  - DHCP v4, IPv4 autoconf, IPv6 SLAAC, DNS, SNTP
- Multiple network interfaces support
- Time Sensitive Networking support
- **BSD Sockets**-based API
- Supports IP offloading
- **Compliance and security** tested



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Bluetooth Low Energy Controller

- **Bluetooth 5.3 compliant** and qualified (5.1)
- Support for multiple BLE radio hardware architectures
- Proprietary radios (downstream only)
- Unlimited role and connection count
- Concurrent multi-protocol support ready
- Multiple advertiser and scanner instances



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Zephyr USB Device Stack

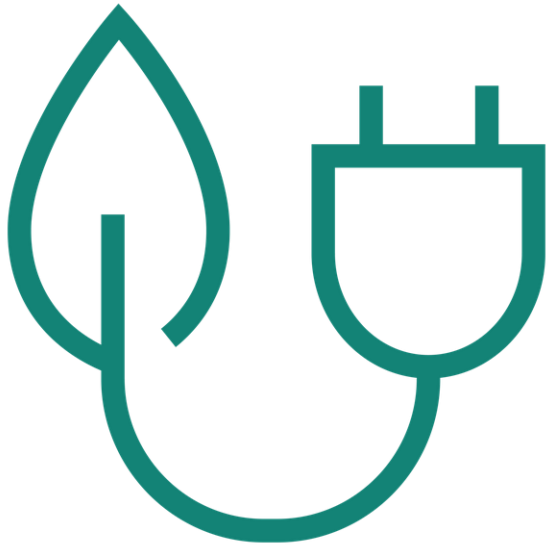
- **USB 2.0** & **USB-C** support
- Supports multiple MCU families (PSOC Edge, STM32, Kinetis, nRF, SAM,…)
- Supports most common devices classes: CDC, Mass Storage, HID, Bluetooth HCI over USB, DFU, USB Audio, etc.
- Tight integration with the RTOS
- Native execution support for emulated development on Linux
- WebUSB support



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Power Management

- Services aimed at reducing power consumption while preserving responsiveness
- **Key concepts**
  - Tickless kernel
  - System PM: idle thread, interruptions only for registered events
  - Device PM: device drivers can react to PM state changes
- Handled by the kernel / Customizable by the user
- Automated testing using Twister Power Harness



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Secure boot / Device Management

- Leverage **MCUboot** as secure bootloader
- Application binary can be signed/encrypted
  - Can use hardware keys
- But also:
  - Downgrade prevention
  - Dependency checks
  - Reset and failure recovery
- Over-the-air (OTA) upgrades
  - OMA LwM2M, Eclipse hawkBit
  - Vendor offerings



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Hardware security

- **Cryptography APIs**
  - Random Number Generation, ciphering, etc.
  - Supported by crypto HW, or SW implementation (TinyCrypt)
- **Trusted Firmware integration**
  - Firmware verification/encryption
  - Device attestation
  - Management of device secrets



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Zephyr Dev Community

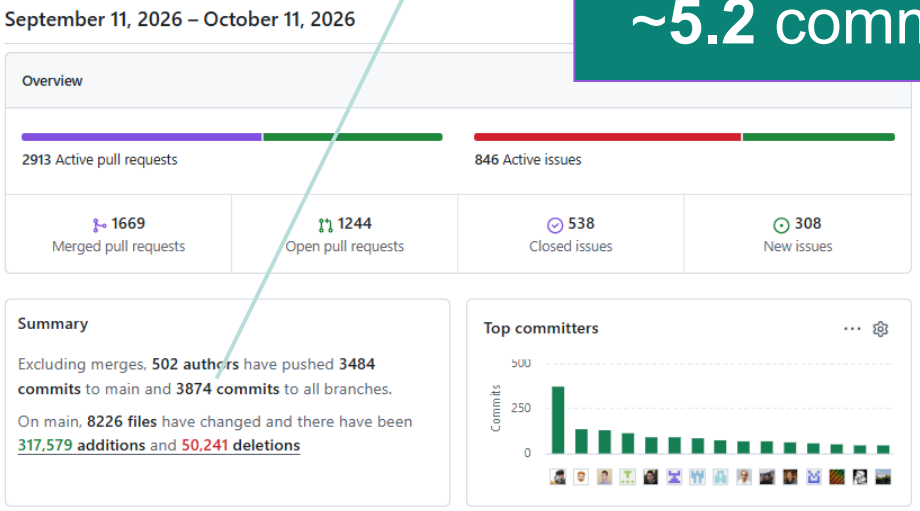
Content in this section is taken from Zephyr marketing collateral

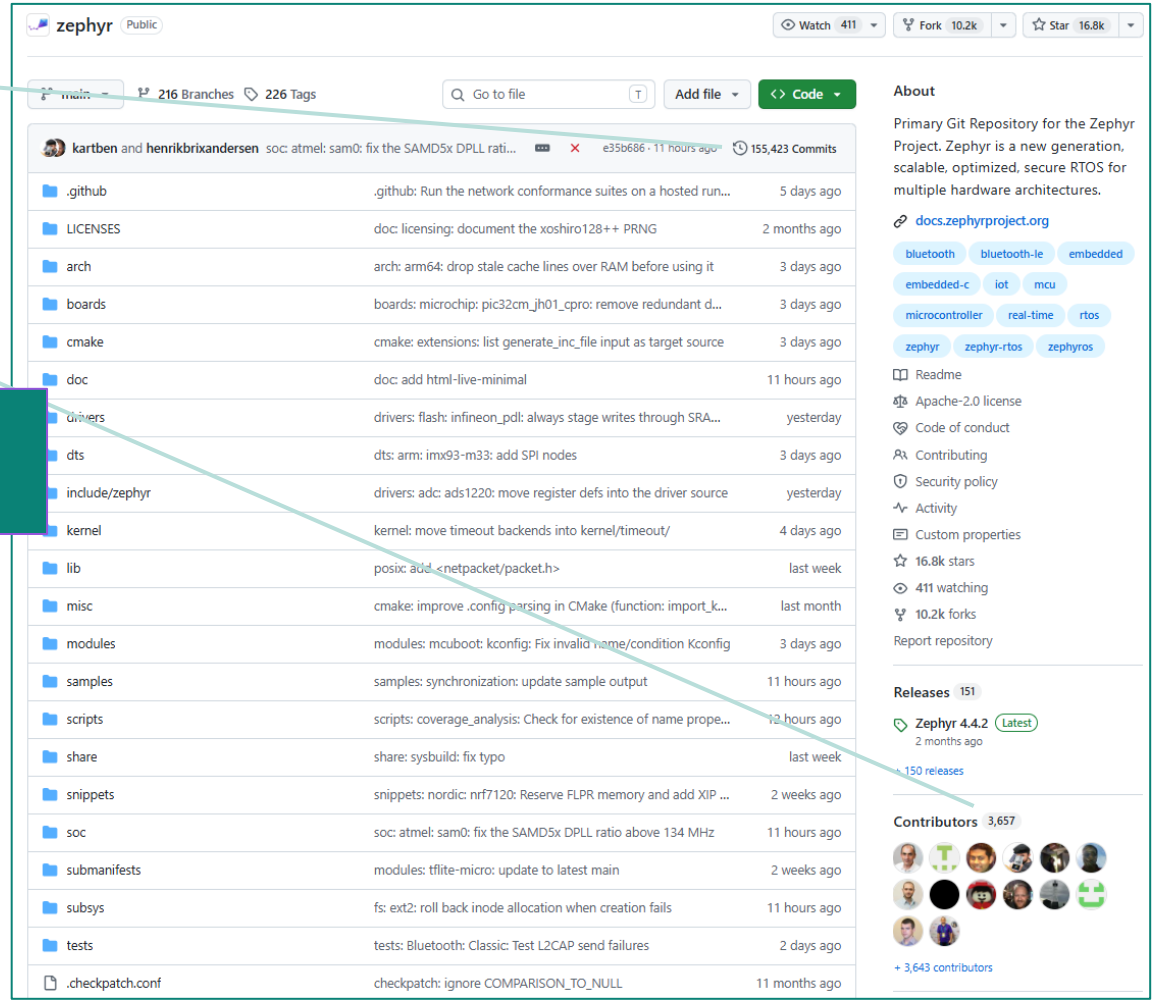
Vibrant and active community of contributors

https://github.com/zephyrproject-rtos/zephyr
Total commits: 155,000+
Total contributors: 3,600+

GitHub stars: 16,754 | Forks: 10,157
Commits in the last 30 days: 3,874
Latest tagged release: v4.4.2 (Aug 2026)

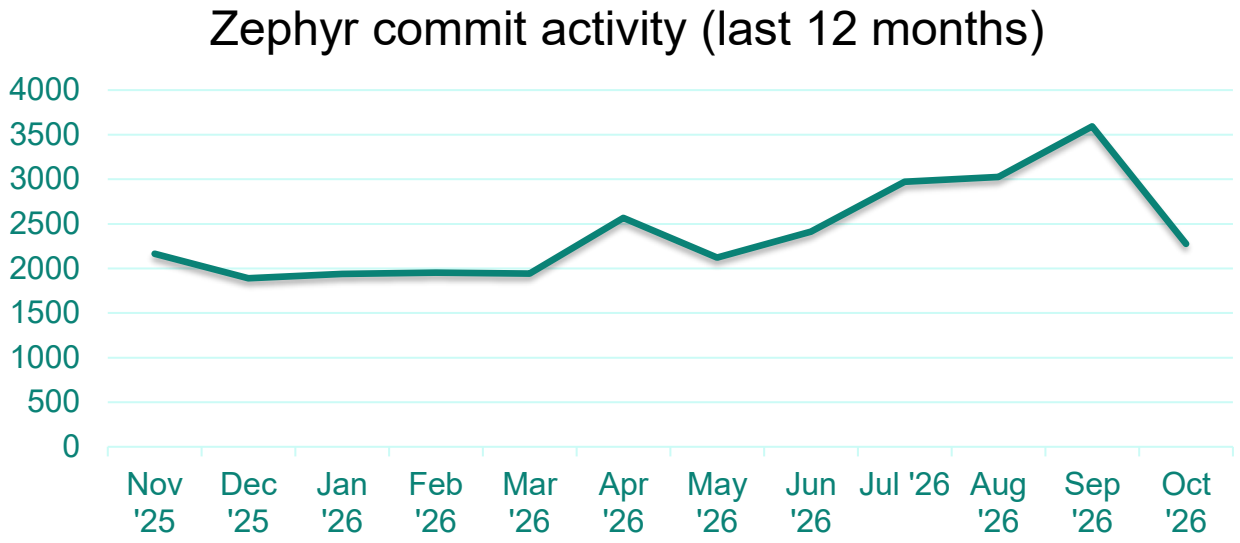
~5.2 commits/hour





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GitHub Community Growth



github.com/zephyrproject-rtos

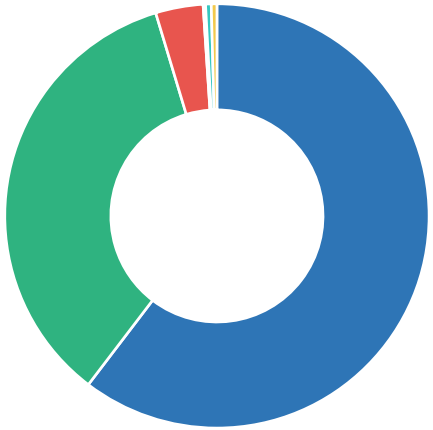
What this data shows:

- Commit activity nearly doubled over the past year
- 16,756 stars and 10,158 forks - a large, broad-reach open-source community.
- Fork-to-star ratio (~61%) is high - many developers are actively building on Zephyr, not just bookmarking it.



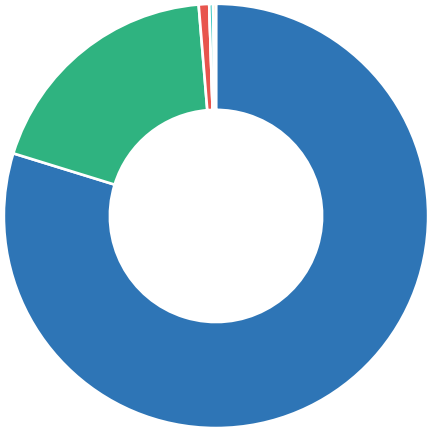
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Share of voice (search & social mentions) vs. other RTOS projects



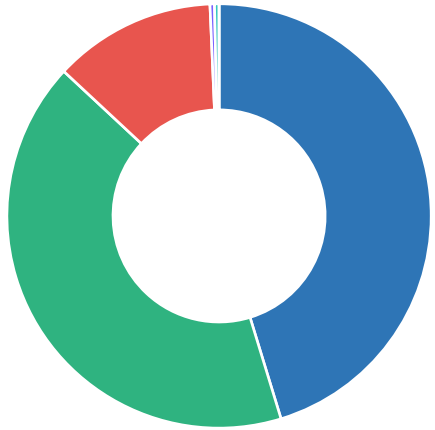
RTOS	Mentions
Zephyr Project	984 (60%)
FreeRTOS	570 (35%)
ThreadX	59 (4%)
ARM Mbed	3 (0%)
RIOT	7 (0%)
NuttX	7 (0%)

January 2026



RTOS	Mentions
Zephyr Project	1.1K (79%)
FreeRTOS	261 (20%)
ThreadX	11 (1%)
ARM Mbed	0 (0%)
RIOT	4 (0%)
NuttX	3 (0%)

February 2026



RTOS	Mentions
Zephyr Project	662 (45%)
FreeRTOS	609 (42%)
ThreadX	181 (12%)
ARM Mbed	5 (0%)
RIOT	5 (0%)
NuttX	0 (0%)

March 2026

Source: Zephyr Project marketing team (search + social listening tool), Q1 2026 snapshot.

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3 Getting started with Zephyr (Development Board Support)

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Infineon's Platinum Membership benefits



Industry leadership



Fast track to Technical Steering Committee Participation



Help shape the Safety & Certification Programs



Marketing Opportunities



Member Networking Opportunities



Learning & Engagement

Zephyr® Project MEMBER

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Infineon supported development boards

20 boards, 7 MCU families - in-tree Zephyr support, newest/most-complete families first

PSOC™ Edge
KIT_PSE84_EVAL - E84 Evaluation Kit
KIT_PSE84_AI - E84 AI Evaluation Kit

PSOC™ Control
KIT_PSC3M6_EVAL - C3M6 Evaluation Kit
KIT_PSC3M5_EVK - C3M5 Evaluation Kit
KIT_PSC3M5_CC2 - C3M5 Connectivity Card 2

AIROC™
CYW920829M2EVK-02 - CYW20829 BLE MCU

PSOC™ 4
CY8CPROTO-041TP - PSOC™ 4100T Plus
CY8CKIT-041S-MAX - PSOC™ 4100S Max
CY8CPROTO-040T - PSOC™ 4000T
CY8CPROTO-040T-AUTO - PSOC™ 4000T AUTO

PSOC™ 6
CY8CKIT-062S2-AI - PSOC™ 6 AI Eval Kit
CY8CPROTO-062-4343W
CY8CKIT-062S4 - PSOC™ 62S4 Pioneer Kit
CY8CPROTO-063-BLE

TRAVEO™ T2G
KIT_T2G-B-H_EVK - Body High Eval Kit
KIT_T2G-B-H-LITE - Body High Lite Eval Kit
KIT_T2G-C2D6M-LITE

XMC™
KIT_XMC72_EVK - XMC7200 Evaluation Kit
XMC47-RELAX-KIT
XMC45-RELAX-KIT













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Infineon targeted support in Zephyr® RTOS Ecosystem



Competitive support
Best-in-Class support

Device Support

- All new MCUs and Connected MCUs include Zephyr support by default at public release, unless explicitly exempted
- All applicable Zephyr driver APIs that map to supported device peripheral are supported
- Enabled Zephyr APIs make best use of Low-Power, DMA, and Security features available in Zephyr
- All mass-market focused development kits are supported directly within Zephyr
- Differentiated device capabilities are enabled using Custom APIs, per marketing/product requirements

Tool Enablement

- Code Generation of device configuration / initialization using graphical configuration tools
- Ease-of-use assets (videos, documentation, tool extensions, installers) targeting bring-up of Zephyr development environment for Infineon devices
- Contribution and/or development of IDE support that fully leverages the Zephyr build environment and enables debugging on Infineon devices
- AI assisted documentation and coding optimized with relevant Zephyr content for Infineon devices

Software Ecosystem

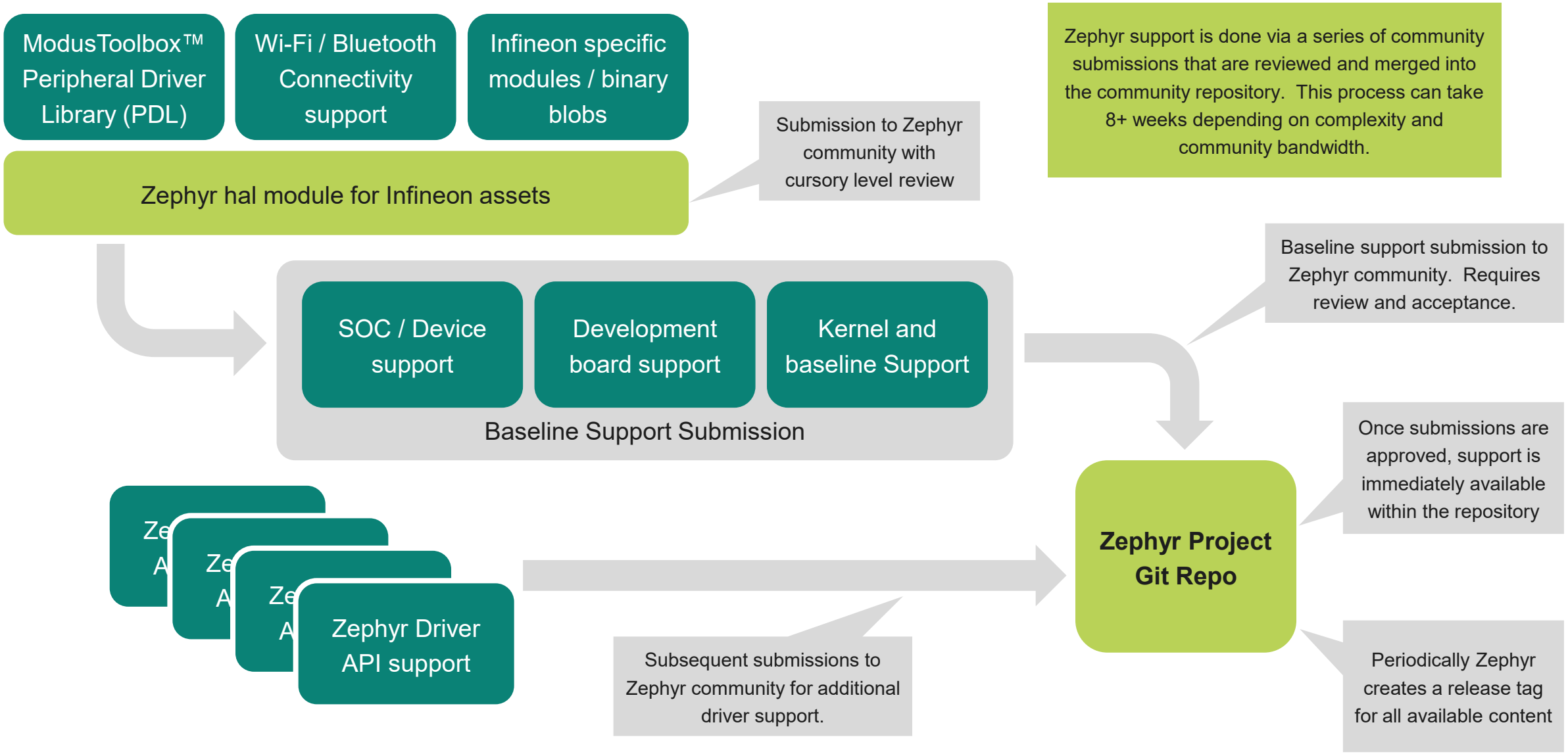
- Support for standard and applicable Zephyr OS Service, including Cryptography, Device Management, Power Management, TF-M
- Support for Device Management DFU/OTA within Zephyr (leveraging MCUBoot)
- Support for standard Zephyr connectivity: Zephyr Bluetooth LE Stack/Controller, Zephyr CAN, Zephyr Networking, Zephyr USB
- Well documented support for out-of-tree middleware targeting key device differentiated software libraries (Apple Find My, CAPSENSE, Safety Test)

Ecosystem Engagement

- Active Platinum level membership in Zephyr engagement
- Increased technical training and expertise across customer support contacts (FAE, AE)
- Active contribution on technical steering committees and engagement with community
- Targeted Zephyr marketing and advocacy efforts (webinars, meet-ups, conferences, whitepapers, events)

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Understanding the Zephyr release process

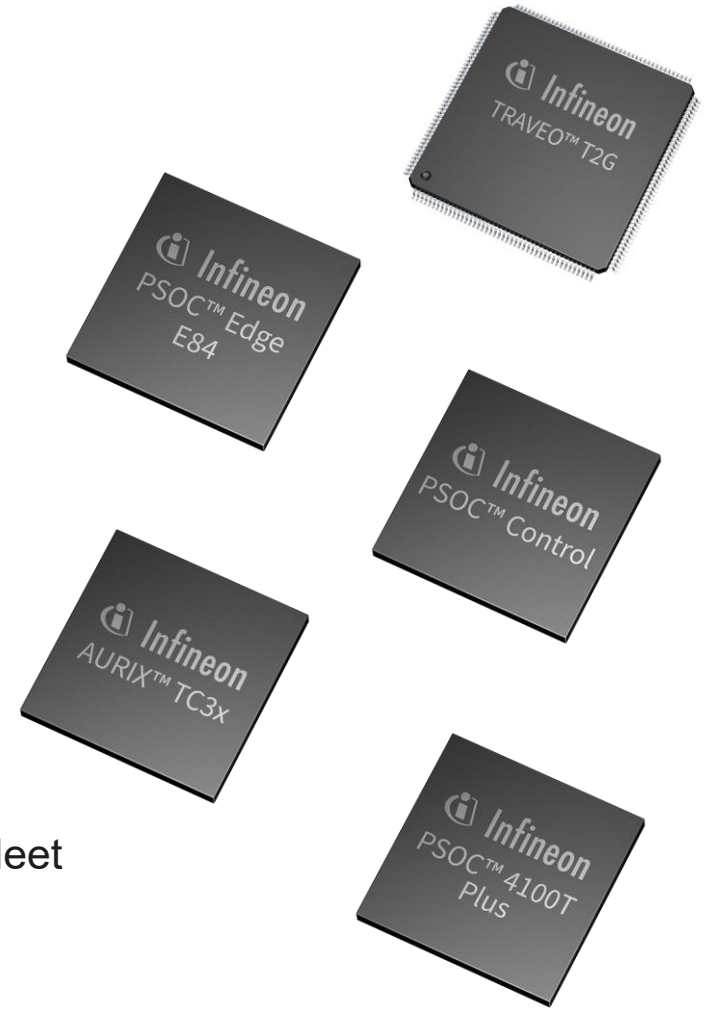


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Infineon Development Roadmap for Zephyr

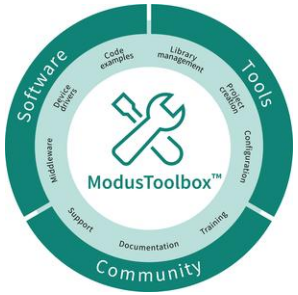
- **Zephyr 4.4.2 released August 2026**
  - Full Device Base support for PSOC™ Edge E84
  - Full Device Base support for PSOC™ Control C3
  - Expanded peripheral coverage: ADC, PWM, CAN FD, CAPSENSE

- **Near-term (Q3-Q4 CY2026)**
  - Continued PSOC™ Control C3 coverage (CAN FD, HPPASS ADC)
  - Safety Test Library (IEC 60730) workflow integration
  - PSE84 multi-core and TF-M / TrustZone security enablement
  - PSOC™ 4 HV family expansion (HVMS, HVPA, 040T-AUTO)
  - CAPSENSE™ Zephyr tooling integration
  - TRAVEO™ / AURIX™ automotive enablement
  - Tool enablement & software ecosystem support across the Infineon Zephyr board fleet



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Understanding ModusToolbox™ and Zephyr



– Provides a targeted ecosystem focused on development of Infineon devices.

– IDE support, including Eclipse IDE, VS Code, and 3rd party IDEs.
– Support for GCC, IAR, and Arm compilers using ‘make’ based ModusToolbox™ build system.
– Graphical UI configurations tool
– Leverages Infineon developed low-level drivers, middleware abstraction, middleware, and code examples.
– Enables key partner middleware via ModusToolbox™ & Friends program

Zephyr®

– Provides a community driven ecosystem with support for a targeted set of MCUs from a wide range of silicon providers.

– Targeted at command-line users with some community support for VS Code.
– Default support for GCC compiler using ‘west’ based Zephyr build environment. Additional toolchains are now being supported.
– Text-based configuration via Kconfig
– Leverages vendor provided device support; Infineon support based on low-level drivers.
– Community contributed middleware, with support for Zephyr defined driver API, stacks, and RTOS.

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Setup Checkpoint - Toolchain & Finishing the Workspace

- Once the download finishes, install the Zephyr SDK:
  - **west sdk install -t arm-zephyr-eabi**

- Picks the right SDK version automatically, installs only the ARM toolchain, and registers itself so every future build finds it

- Finish with three short commands:
  - **pip install -r devcon-training-2026/requirements-devcon.txt**
  - **west config build.cmake-args -- "-DOPENOCD=<Infineon OpenOCD path>"**
  - **west patch apply** (graphics-lab fixes)

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Table of contents

1 Zephyr Project Introduction (Overview | Ecosystem | Dev Community)

2 Infineon's involvement within Zephyr (Platinum Membership | TSC)

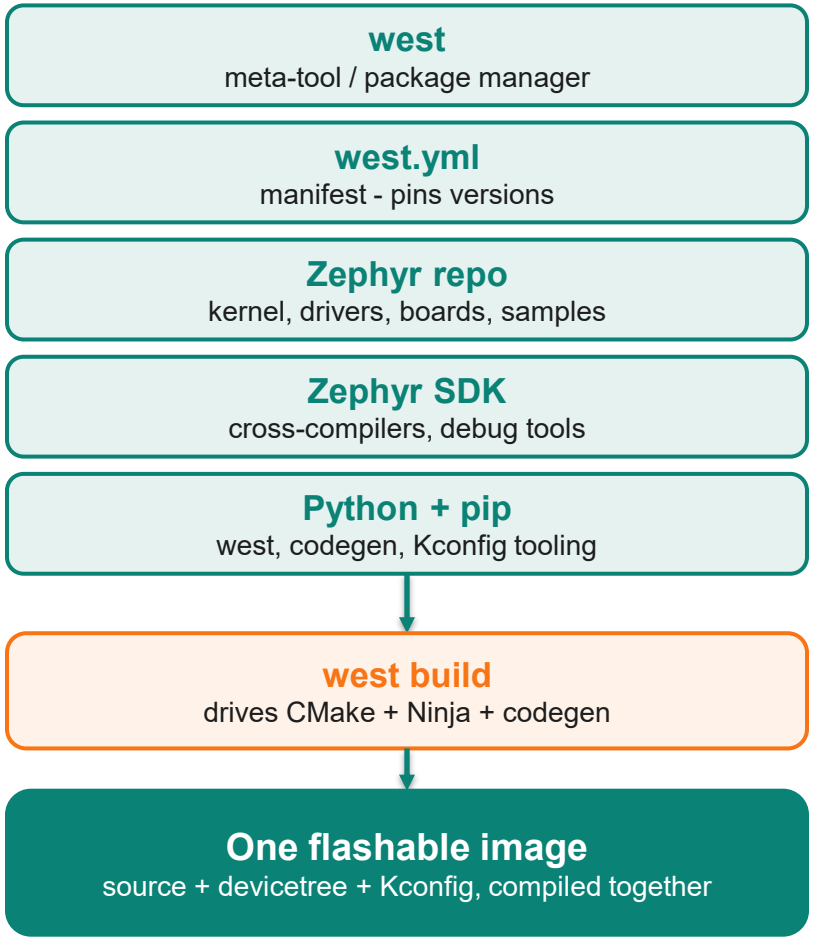
3 **Getting started with Zephyr (Development Board Support)**

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The Pieces of Your Dev Environment

- **west** - meta-tool & package manager
- **west.yml** - manifest, pins every version
- **Zephyr repo** - the RTOS source itself
- **Zephyr SDK** - cross-compilers & debug tools (separate install)
- **Python + pip** - runs west, Kconfig, devicetree tooling

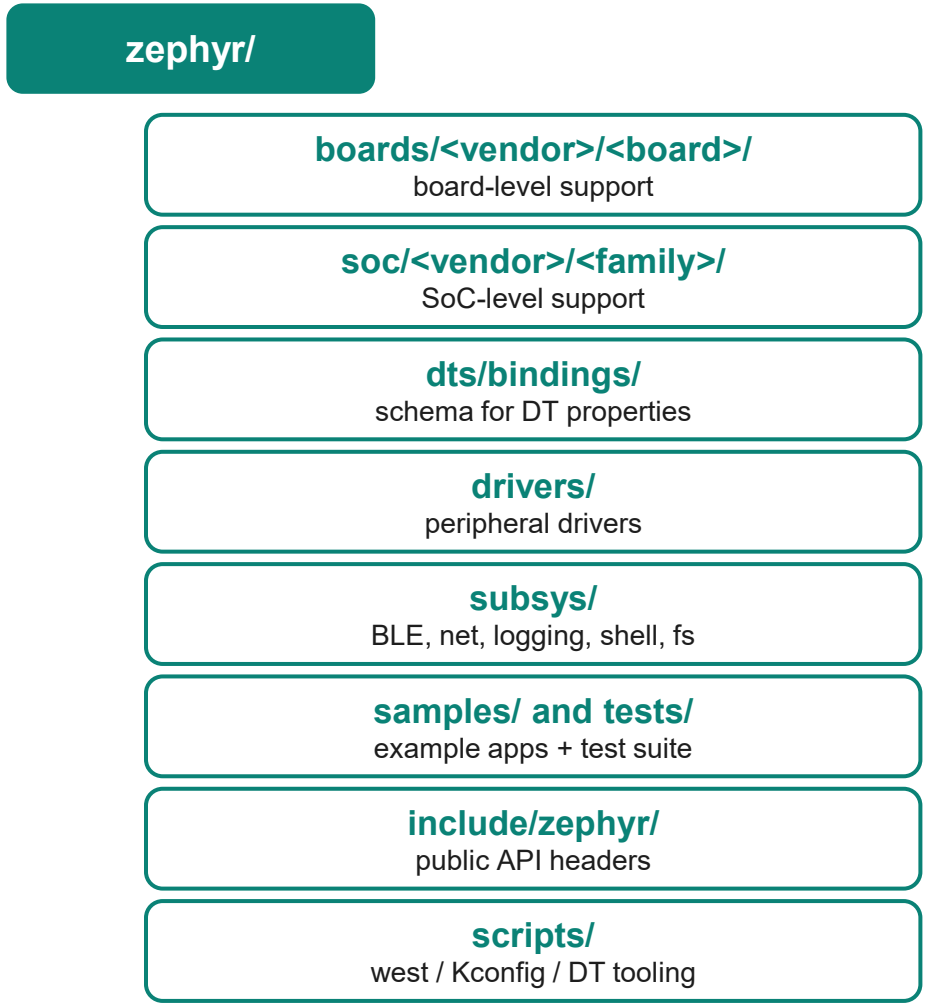
- **west build → CMake + Ninja → one flashable image**



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Zephyr Repository - Where Things Live

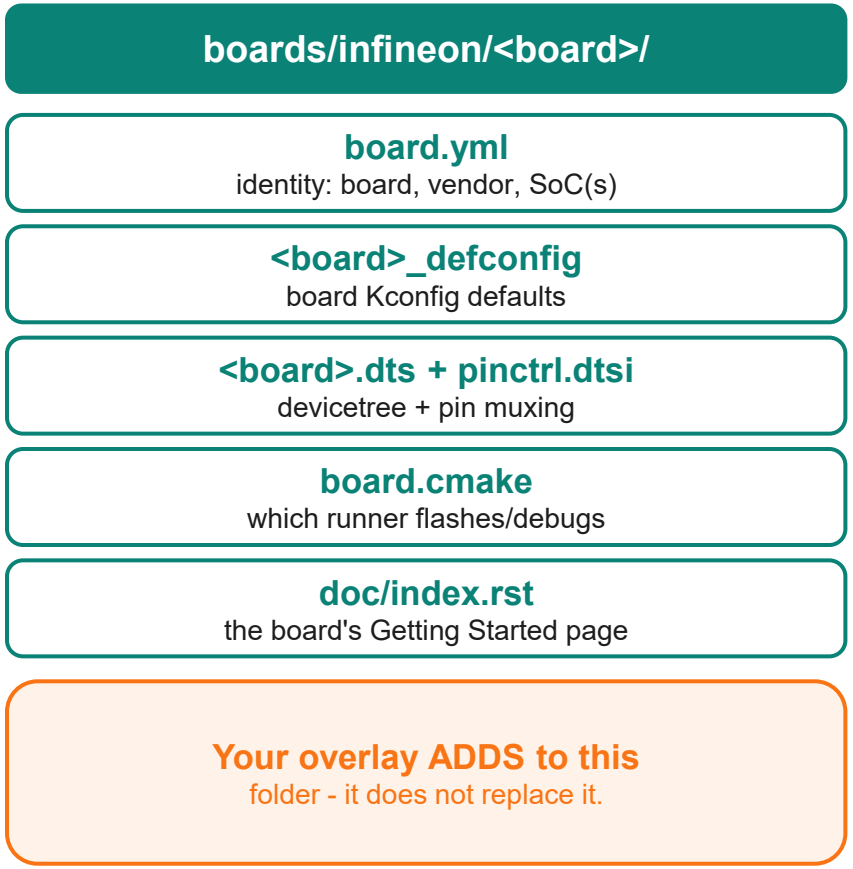
- **boards/<vendor>/<board>/** - board-level support
- **soc/<vendor>/<family>/** - SoC-level support
- **dts/bindings/** - devicetree property schema
- **drivers/** - peripheral drivers
- **subsys/** - BLE, networking, shell, filesystems
- **samples/ and tests/** - ready-to-build examples
- **include/zephyr/** - public API headers
- **scripts/** - west / Kconfig / DT tooling



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Anatomy of a Board Support Package (BSP)

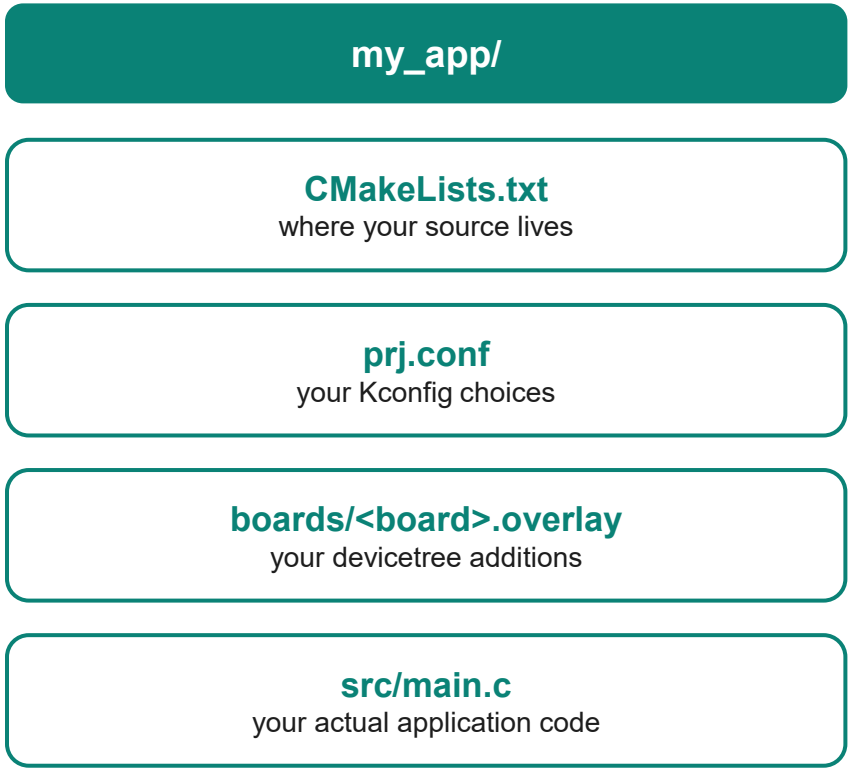
- Same file set, every board
- **board.yml** - identity (board, vendor, SoC)
- **<board>_defconfig** - board Kconfig defaults
- **<board>.dts + pinctrl.dtsi** - devicetree + pin muxing
- **board.cmake** - flash/debug runner
- **doc/index.rst** - Getting Started page
- **Your overlay ADDS to this, doesn't replace it**



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Anatomy of a Zephyr Application

- **CMakeLists.txt** - where your source lives
- **prj.conf** - your Kconfig choices
- **boards/<board>.overlay** - your devicetree additions
- **src/** - your application code
- **Same shape, every app, every board**



Same four-file shape, every app, every board -
nothing here is Infineon-specific.

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Sample Applications - A Guided Tour

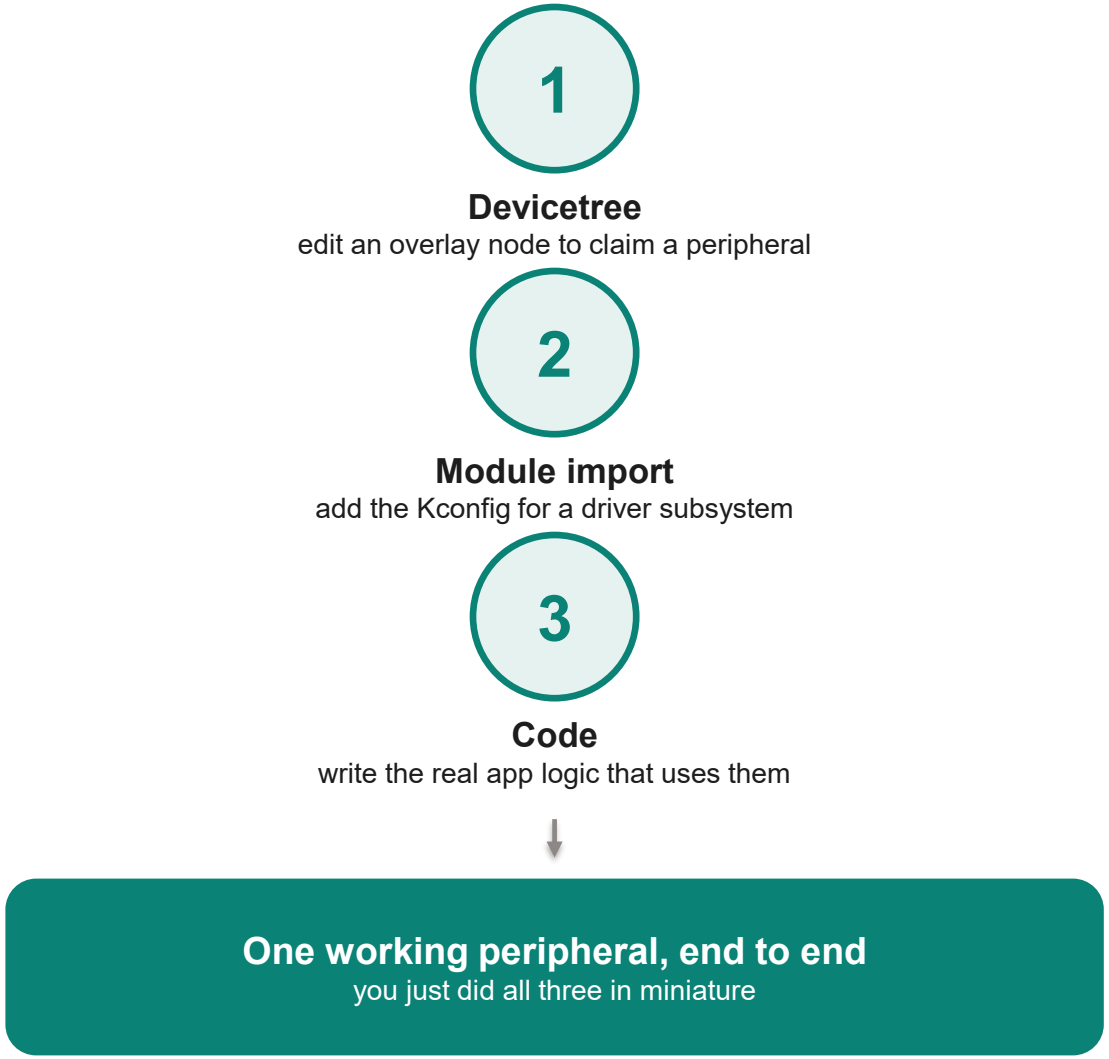
- **hello_world, basic/*** - zero/low-dependency smoke tests (blinky, button, fade_led)
- **subsys/shell/shell_module** - interactive console shell
- **drivers/*** - 70+ peripheral demos: adc, can, pwm, i2c, spi, watchdog, counter...
- **sensor/*** - ~75 vendor sensor demos, plus sensor_shell for any supported sensor
- **subsys/*** - usb, usb_c, fs, settings, zbus, modbus, canbus, logging
- **bluetooth/*** - ~90 BLE central/peripheral/mesh samples
- **net/*** - sockets, MQTT, CoAP, HTTP client/server
- **boards/infineon/** - Infineon's own in-tree sample (autanalog)

- **Check sample.yaml → platform_allow for board support**

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The Three Touchpoints – Preview of upcoming labs

1. **Devicetree** - edit overlay node
2. **Module import** - add Kconfig
3. **Code** - write real app logic



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Devicetree - Your First Touchpoint

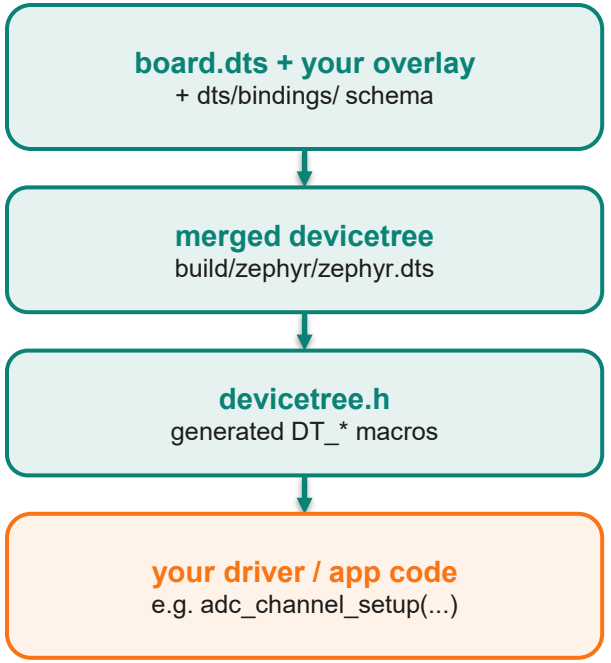
- **Board** = what hardware exists
- **Overlay** = what your app uses

- Reference from PSOC™ Control CAN lab:

```
zephyr,user {
    // TODO 1a: uncomment to claim the pot's ADC channel
    io-channels = <&adc0 12>;
};

aliases {
    // TODO 1b: uncomment to alias the PWM LED node
    cmd-pwm = &pwm_cmd;
};

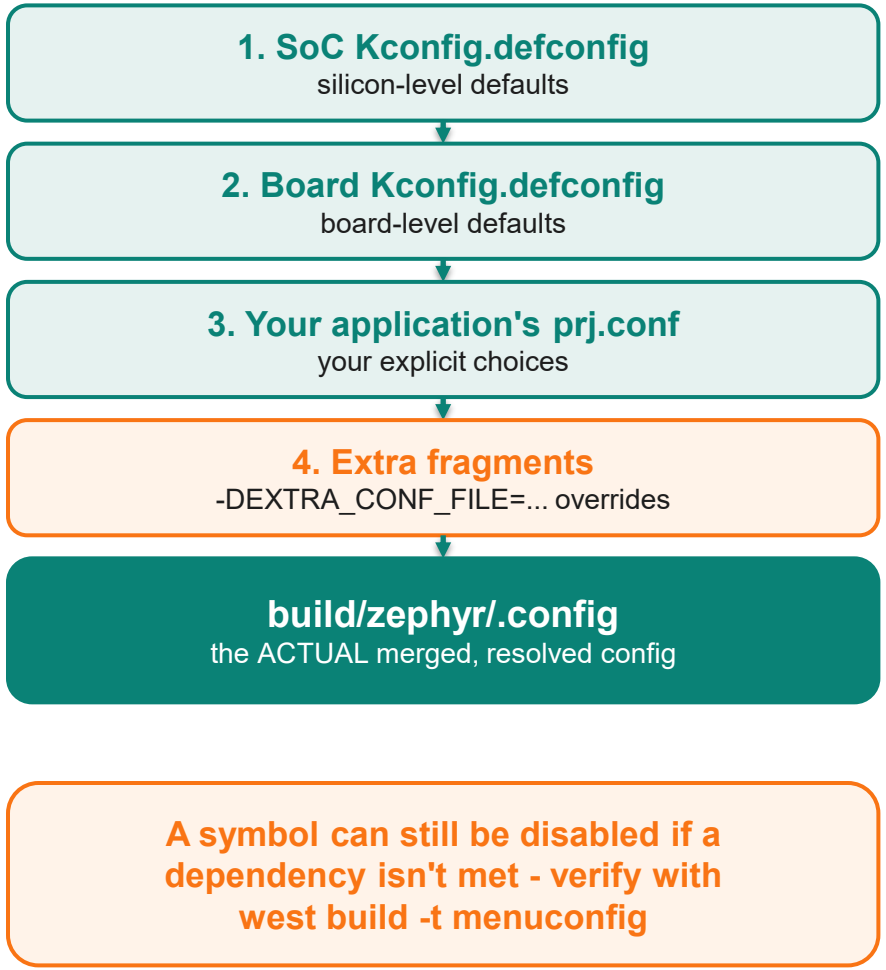
// TODO 1b: uncomment this whole pwmleds block
pwmleds { compatible = "pwm-leds"; ... };
```



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Kconfig - Your Second Touchpoint

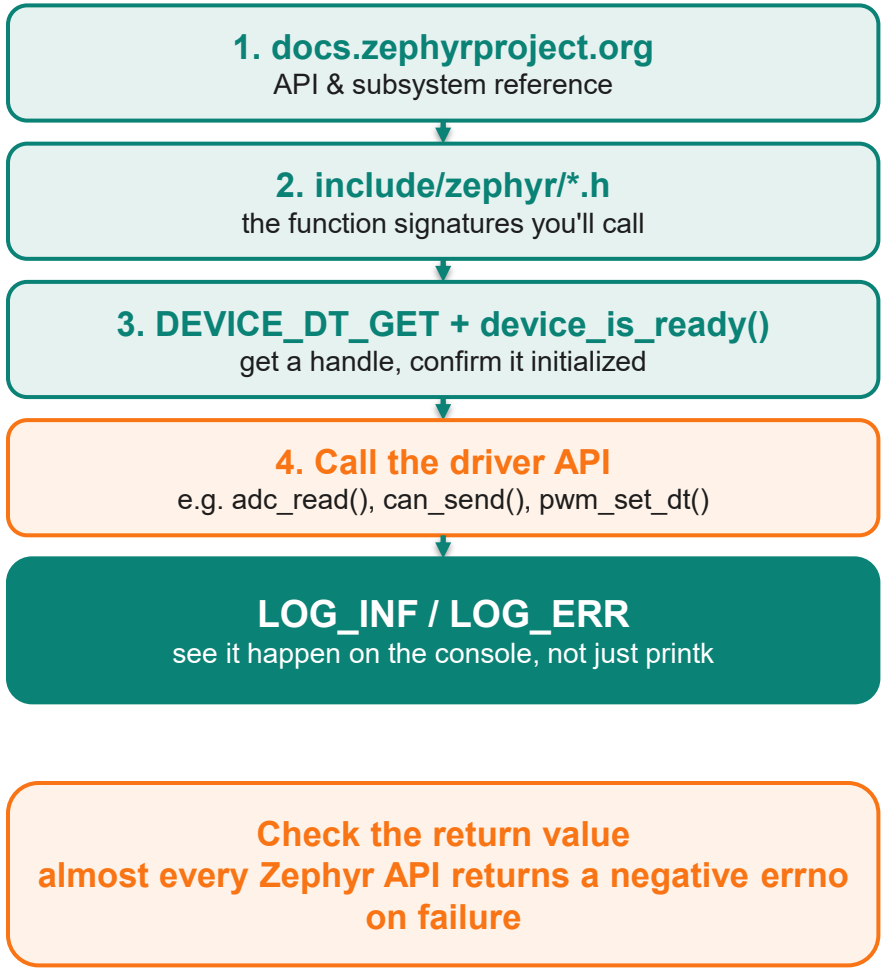
- **Kconfig** = which subsystems get compiled in
- **prj.conf** = your choices, merged with board defaults
- Examples from this week's lab:
  - CONFIG_CAN=y
  - CONFIG_ADC=y
  - CONFIG_PWM=y
- **menuconfig / guiconfig** - browse & search interactively
- **Merge order:**
  SoC → board → prj.conf → extra fragments
- **Verify** build/zephyr/.config, not just prj.conf
- Use **west build -t menuconfig** to check



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Code - Your Third Touchpoint

- **API docs**: docs.zephyrproject.org/latest – full Doxygen reference
- **Headers**: include/zephyr/<subsystem>/*.h – the real function signatures
- Pattern from this week's lab:
  - DEVICE_DT_GET(DT_CHOSEN(zephyr_canbus))
  - device_is_ready(can_dev)
  - LOG_ERR("CAN device not ready")
- **device_is_ready()** - confirms the driver actually initialized, before you touch it
- **Logging: LOG_INF / LOG_WRN / LOG_ERR** instead of printk for anything you ship
- **Check** the return value - almost every Zephyr API returns a negative errno on failure
- A dropped return code: **Zephyr's #1 time sink**



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Key reference links for Zephyr and Infineon MCU



[Infineon Zephyr RTOS Product Page]



[Zephyr RTOS GitHub Repository]



[Infineon Zephyr RTOS Discord Channel]



Getting Started with:

CY8CPROTO-041TP (PSOC™ 4100T Plus)

KIT_PSC3M6_EVAL (PSOC™ Control C3M6)

KIT_PSE84_EVAL (PSOC™ Edge E84)



[PSOC™ 6 and Zephyr Video Tutorial]

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Build & Flash Blinky - Prove the Chain Works

Your kit	Board target	Clean build time
CY8CPROTO-041TP (PSOC™ 4100T Plus)	cy8cproto_041tp	~2 minutes
KIT_PSC3M6_EVAL (PSOC™ Control C3M6)	kit_psc3m6_evk	~1.5 minutes
KIT_PSE84_EVAL (PSOC™ Edge E84)	kit_pse84_eval/pse846gps2dbzc4a/m55	~4 minutes (needs --sysbuild)

west build -b <board> -d build/blinky -s zephyr/samples/basic/blinky

PSOC™ Edge needs --sysbuild (for CM55 builds) - the build succeeds, but without it nothing starts the core

west flash -d build/blinky

Success = a memory usage report ending the build, and a blinking LED

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If Something Goes Wrong

- 'west' is not recognized → reopen your terminal; pip install west did not land on PATH
- west: unknown command "sdk" → west update has not finished, or you're not inside devcon-ws
- ERROR: board <name> not found → typo in the board target, or running from outside devcon-ws
- Build succeeds, west flash can't find OpenOCD → the west config line from Setup is missing or wrong
- Build succeeds, flash succeeds, no LED (PSOC™ Edge) → you left out --sysbuild
- west update stops partway → network issue; run west update --narrow again, it resumes

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Working in VS Code with Zephyr

- west on the command line is the one skill every attendee needs - but a good editor still earns its keep:
  - Syntax highlighting for devicetree (.overlay) and Kconfig (prj.conf)
  - One-click build/flash/debug instead of retyping commands

- Two supported paths today - pick based on how much you want VS Code to manage for you:
  - Option A - Ac6 Zephyr Workbench: one extension, manages SDK/toolchain/build/flash/debug
  - Option B - lightweight extensions: better editing only, you keep driving west yourself

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Option A - Ac6 Zephyr Workbench (full IDE)

- Install Ac6.zephyr-workbench from the VS Code Marketplace
- Open your devcon-ws folder - it auto-detects the workspace, Zephyr SDK, and Infineon boards, no manual import needed
- Validated in hands-on testing:
  - Infineon boards appear in the board picker out of the box (no BOARD_ROOT config needed)
  - Debug-tools installer fetches Infineon's real ModusToolbox™ Programming Tools - not a reimplementation
  - Good clangd defaults: auto-wires the Zephyr SDK toolchain, disables the conflicting default IntelliSense engine
- Known rough edges to plan for:
  - OpenOCD/J-Link installers need UAC admin rights - may be blocked on locked-down laptops
  - No bundled Kconfig support - add luveti.kconfig separately (not trond-snekvik.kconfig, which is Nordic-branded)
  - A per-app venv path bug is non-fatal only if you already have system Python on PATH

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Option B - Lightweight Extensions-Only Path

- For attendees comfortable with west on the command line who just want a better editor

- Install these three extensions:
  - clangd - IntelliSense (disable the default C/C++ extension's engine to avoid conflicts)
  - KyleMicallefBonnici.dts-lsp - devicetree/.overlay syntax highlighting
  - luveti.kconfig - Kconfig syntax highlighting (vendor-neutral)

- Wrap west build / west flash in a simple VS Code task (tasks.json) if you want a button instead of typing commands

- No SDK/toolchain management, no board auto-detection - you're still driving west yourself, just with a nicer editor

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